

Johannes Hosle

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EMPLOYMENT

Carnegie Mellon University in Qatar Postdoctoral Fellow in Mathematics Mentors: Layan El Hajj and Henrik Shahgholian	2026-present
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EDUCATION

Massachusetts Institute of Technology Ph.D. in Mathematics Advisor: David Jerison	2020-2026
University of California, Los Angeles B.Sc. in Mathematics with Departmental Honors, summa cum laude M.A. in Mathematics	2016-2020

PUBLICATIONS AND PREPRINTS

J. Hosle, Monotonicity formulas in positively curved settings with applications to two-phase free boundary problems, in preparation
J. Hosle, The isomorphic section-projection problem for convex bodies , submitted
J. Hosle and P. Ivanisvili, Geometric Block Exponents and a Uniform Mixed-Alphabet Sumset Inequality , submitted
J. Hosle, Stolarsky-Type Inequalities in a Max-Convolution Problem , submitted
J. Hosle, A. V. Kolesnikov, and G. V. Livshyts, On the Lp-Brunn-Minkowski and dimensional Brunn-Minkowski conjectures for log-concave measures, J. Geom. Anal. (2021)
J. Hosle, On extensions of the Loomis-Whitney inequality and Ball's inequality for concave, homogeneous measures, Adv. Appl. Math. (2020)
J. Hosle, On the Comparison of Measures of Convex Bodies via Projections and Sections, Int. Math. Res. Not. IMRN (2021)

HONORS AND AWARDS

Sigma Xi	2026
Dean of Science Fellowship, MIT	2020-2023
Phi Beta Kappa	2020
Sherwood Prize, UCLA	2020
Dean's Honors List	2016-2020
Putnam Competition, Top 250	2016
National Merit Scholarship Semifinalist	2015

TEACHING AND GRADING

18.095 Mathematics Lecture Series, MIT Teaching Assistant (J. Dunkel)	Jan. 2026
18.065 Matrix Methods in Data Analysis, Signal Processing, and Machine Learning, MIT Teaching Assistant (Z. Chen)	Spring 2025
18.100B Real Analysis, MIT Course Assistant (L. Guth)	Fall 2024
18.100P Real Analysis, MIT Recitation Leader (K. Naff)	Spring 2024
18.01 Calculus, MIT Recitation Leader (L. Guth)	Fall 2023

MENTORSHIP EXPERIENCE

Academic Mentor, Mathroots @ MIT Two-week mathematical talent accelerator for high school students hosted by MIT PRIMES.	July 2026
MIT Directed Reading Program Mentored undergraduates Ryan Dong and Caleb Pascale in convex geometry, including the Busemann-Petty problem and Bourgain's bound for the isotropic constant.	Jan. 2025

PRESENTATIONS

Analysis Seminar, Columbia University Upcoming	Sept. 2026
Analysis Seminar, Brown University Upcoming	Sept. 2026
Synergies between Geometry, Probability, and Computation in High Dimensions ICERM, Brown University, Providence, Rhode Island	Aug. 2026
Workshop in PDE and Applied Math in the Northeast Region University of Connecticut, Storrs, Connecticut	Mar. 2026
Pure Math Graduate Student Seminar, MIT	Feb. 2025
BIRS Workshop: Geometric Tomography BIRS, Banff, Canada	Feb. 2020
Workshop in Convexity and Geometric Aspects of Harmonic Analysis Georgia Institute of Technology, Atlanta, Georgia	Dec. 2019

RESEARCH PROGRAMS AND SUMMER SCHOOLS

Interactions between Convex Geometry and Spectral Analysis Universite de Montreal, Montreal, Canada	July 2025
Particle Interactive Systems: Analysis and Computational Methods SLMath, Berkeley, California	June 2024
Georgia Tech REU in Mathematics Supervisors: Michael Lacey and Galyna Livshyts	Summer 2018

PROGRAMMING

Python: Proficient; background in statistical learning theory; experience with NumPy and scikit-learn.
Julia: Proficient; used for gradient flows in free boundary problems, supersolution construction, and numerical estimates for weighted Wirtinger inequalities.
Wolfram Language: Intermediate; familiar with machine learning methods.
LaTeX: Proficient.

LANGUAGES AND CITIZENSHIP

Languages: English (native), Russian (elementary).
Citizenships: United States and Germany.